

Lesson 14 — DAX Optimization

Topic: DAX Optimization

Prerequisites:

Download Chocolate Sales.csv file.

In Power BI:

Go to File → Options and Settings → Options → Data Load

Disable Auto Date/Time for new files

Create a Date Table using DAX:

Date Table =

```
ADDCOLUMNS(  
    CALENDARAUTO(),  
    "Year", YEAR([Date]),  
    "Month", FORMAT([Date], "MMM"),  
    "Month Number", MONTH([Date])  
)
```

 Learning Levels: Basic → Intermediate → Advanced (1–10)

 Use variable to calculate % Growth in Sales Compared to Last Year

% Growth vs Last Year =

VAR CurrentYearSales =

```
CALCULATE([Total Sales], YEAR('Chocolate Sales'[Date]) = YEAR(TODAY()))
```

VAR LastYearSales =

```
CALCULATE([Total Sales], YEAR('Chocolate Sales'[Date]) = YEAR(TODAY()) - 1)
```

RETURN

```
DIVIDE(CurrentYearSales - LastYearSales, LastYearSales, 0)
```

☑ Calculates year-over-year (YoY) percentage growth.

2 Use variable to calculate difference between current month and previous month

Sales Diff (This vs Prev Month) =

VAR CurrentMonthSales =

```
CALCULATE(  
    [Total Sales],  
    DATESINPERIOD('Chocolate Sales'[Date], MAX('Chocolate Sales'[Date]), -1, MONTH)  
)
```

VAR PrevMonthSales =

```
CALCULATE(  
    [Total Sales],  
    DATESINPERIOD('Chocolate Sales'[Date], MAX('Chocolate Sales'[Date]) - 1, -1, MONTH)  
)
```

RETURN

CurrentMonthSales - PrevMonthSales

☑ Computes month-to-month sales change.

3 Calculate total boxes shipped and average monthly boxes in one measure using VAR

Boxes Summary =

VAR TotalBoxes = SUM('Chocolate Sales'[Boxes Shipped])

VAR AvgMonthlyBoxes =

```
AVERAGEX(  
    VALUES('Date Table'[Month]),  
    CALCULATE(SUM('Chocolate Sales'[Boxes Shipped]))  
)
```

RETURN

"Total Boxes: " & FORMAT(TotalBoxes, "#,##0") &

" | Avg Monthly: " & FORMAT(AvgMonthlyBoxes, "#,##0")

☒ Displays both total and average boxes together.

4) Return only the average monthly boxes

Average Monthly Boxes =

VAR TotalBoxes = SUM('Chocolate Sales'[Boxes Shipped])

VAR MonthsCount = DISTINCTCOUNT('Date Table'[Month])

RETURN

DIVIDE(TotalBoxes, MonthsCount, 0)

☒ Returns only the numeric average.

5) Calculate growth percentage from last month

% Growth from Last Month =

VAR CurrentMonthSales =

CALCULATE(
[Total Sales],
DATESINPERIOD('Date Table'[Date], MAX('Date Table'[Date]), -1, MONTH)
)

VAR PrevMonthSales =

CALCULATE(
[Total Sales],
DATESINPERIOD('Date Table'[Date], MAX('Date Table'[Date]) - 1, -1, MONTH)
)

RETURN

DIVIDE(CurrentMonthSales - PrevMonthSales, PrevMonthSales, 0)

☒ Measures growth compared to previous month dynamically.

6 Create a moving average of sales over the last 3 months

3-Month Moving Avg =

VAR Last3Months =

DATESINPERIOD('Date Table'[Date], MAX('Date Table'[Date]), -3, MONTH)

RETURN

AVERAGEX(Last3Months, [Total Sales])

☒ Smooths out monthly variations and shows trend.

7 Dynamic Card Message (based on Rank and YoY)

Sales Performance Message =

VAR Product = SELECTEDVALUE('Chocolate Sales'[Product Name])

VAR SalesRank =

RANKX(
ALL('Chocolate Sales'[Product Name]),
[Total Sales],
,
DESC,
DENSE
)

VAR SalesYoY =

CALCULATE(
DIVIDE(
[Total Sales] -
CALCULATE([Total Sales], DATEADD('Date Table'[Date], -1, YEAR)),
CALCULATE([Total Sales], DATEADD('Date Table'[Date], -1, YEAR))
)
)

RETURN

SWITCH(
TRUE(),

```

SalesRank <= 3 && SalesYoY > 0.1, "Top Performer - Sales up by " & FORMAT(SalesYoY, "0%"),
SalesRank > 3 && SalesYoY > 0, "Consistent Performer",
"Needs Improvement"
)

```

✓ Use in a Card visual for each product — fully dynamic feedback message.

8 Top 5 Tips to Optimize DAX Queries Manually

#	Optimization Tip	Why It Matters
1	Use VAR to store calculations	Prevents re-computation, speeds up measures
2	Avoid unnecessary iterators like SUMX	Iterators are slow on large datasets
3	Use CALCULATE + FILTER wisely	Simplifies context handling
4	Limit high-cardinality columns	Improves compression and performance
5	Use measure references	Keeps formulas modular and efficient

9 Benefits of DAX Optimization Tools

Tool	Purpose	Benefit
DAX Studio	Query analyzer for DAX	Shows query plan, timings, and storage performance
Performance Analyzer	Built-in Power BI tool	Identifies slow visuals and DAX bottlenecks
Tabular Editor	Model editing & scripting	Allows measure management, bulk edits, and best practices

✓ These tools make DAX models faster, cleaner, and easier to debug.

10 Create a Flag (Top 5 Products by Total Sales)

Top 5 Product Flag =

VAR ProductRank =

```

RANKX(
    ALL('Chocolate Sales'[Product Name]),
    [Total Sales],
    ,
    DESC,

```

```
DENSE
)
RETURN
IF(ProductRank <= 5, "Yes", "No")
```

✓ Returns "Yes" if a product is in the Top 5 by Total Sales.

Summary

By completing this lesson, you can:

Optimize DAX queries using variables and context.

Build time intelligence measures efficiently.

Use ranking, YoY growth, and dynamic cards for insights.

Apply professional optimization tools for real-world performance improvement.